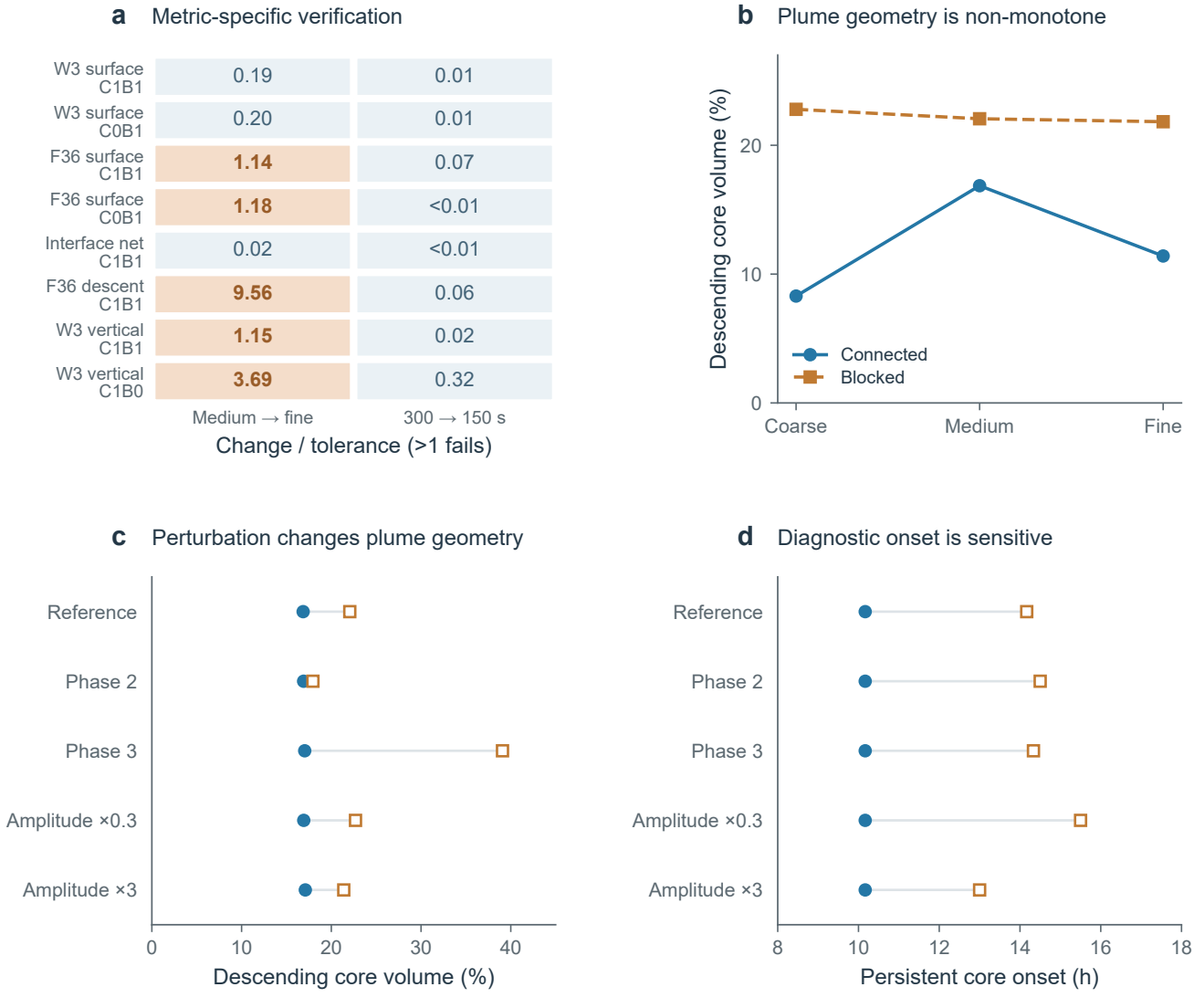

a: tolerance ratios, rounded; amber cells fail. b–d: B1, core-descent criterion (Darcy speed >1e-8 m/s). Surface tolerance 2%; flux/volume tolerance 5%. Onset is a diagnostic, not an instability threshold.